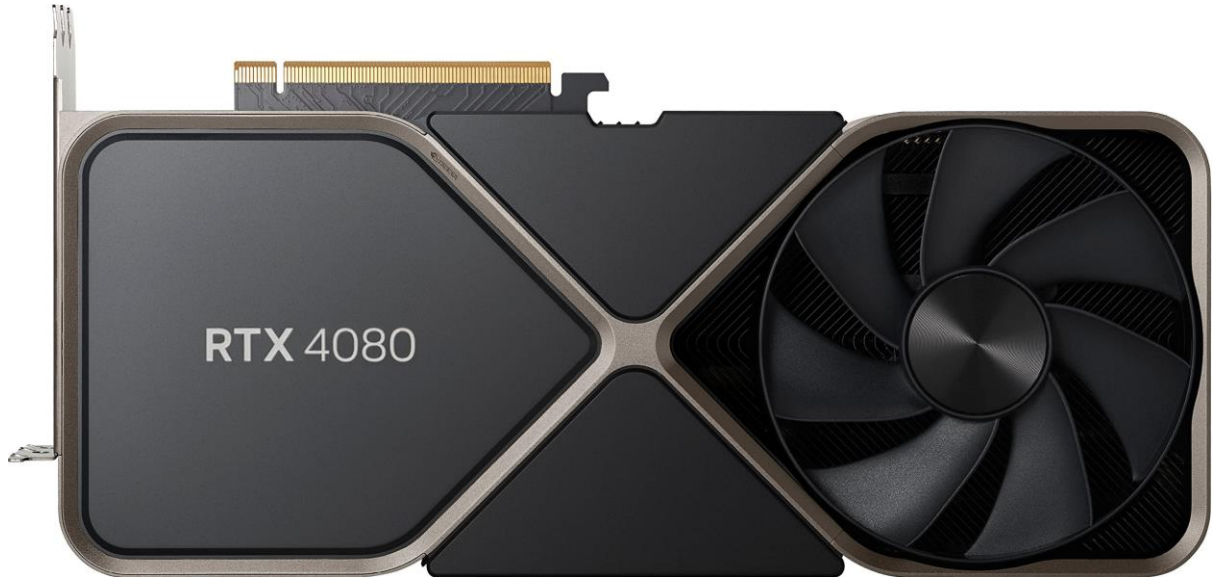


1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.



RTX 4080

Brand: NVIDIA

Model Number: 900-1G136-2560-000

Dedicated

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



A CPU cooling system effectively moves heat away from sensitive (and expensive) component parts through the use of heat pipes and heat sinks. Heat pipes are rapid heat-transferring devices with high thermal conductivity, sometimes up to 100 times more effective at conducting heat than typical metals.

3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling(GPU)	Liquid Cooling(GPU)
Cost	Generally less expensive, requires only heatsinks and fans	More expensive due to radiators, tubing and water blocks.
Maintenance	Low maintenance	High maintenance
Efficiency	Effective for most GPUs but it can struggle with high heat loads	Provides better heat dissipation and low temperature

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.
- I would choose liquid cooling for the GPU. Liquid cooling generally keeps GPU temperatures lower under sustained heavy workloads, which is useful when gaming and streaming simultaneously because both activities place significant demands on the system. Lower temperatures can help maintain higher boost clocks and reduce thermal throttling, leading to more consistent performance